

Ken ZOU

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EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL)

Starting in Sept. 2026

M.Sc. in Quantum Science & Engineering | Master Excellence Fellow

Lausanne, Switzerland

- Coursework: Computational Methods for Molecules, Quantum Simulations of Materials, Molecular Quantum Dynamics.

ESPCI Paris – PSL | PSL University

Sept. 2023 – Aug. 2026

Grande École Engineering Degree in PChem | GPA: 4.0/4.0 | Rank: 1/88 (2024), 2/106 (2025)

Paris, France

- Specialization: Waves and Light-Matter Interactions, Relativistic and Quantum Engineering, Magnetism and Superconductivity, Advanced Condensed Matter Physics, Synthetic Chemistry.
- Teaching Assistant | First-year pre-entry course in Organic Chemistry (15+ hours).
- Representative of ESPCI's Cohort 142 Student Council | Acted as liaison between the administration and Cohort 142 on matters relating to the instruction delivered to engineering students. Prepared a consolidated report on course satisfaction surveys.

Lycée Louis-Le-Grand (LLG)

Sept. 2021 – July 2023

Intensive Undergraduate Preparatory Program in Maths, Physics and Chemistry | GPA: 4.0/4.0

Paris, France

- Advanced oral and written coursework problem-solving in pure applied mathematics, theoretical & experimental physics and chemistry, and computer science in the best French university-level formation for *Grandes Écoles*: École Polytechnique, ENS Paris – PSL, Mines Paris – PSL, Centrale Supélec, ESPCI Paris – PSL.

RESEARCH EXPERIENCE

University of California, Berkeley | College of Chemistry

Apr. 2026 – Aug. 2026

Visiting Student Researcher – Zuerch Lab | Advisor: Michael W. Zuerch

Berkeley, CA

- Designed and ran a cryogenic MIR-pump/sHHG-probe experiment on NiPS₃, revealing two distinct subsystems (spin vs. phonon) via a symmetry-breaking signature in even/odd harmonics. Isolated a slow spin-lattice relaxation restoring antiferromagnetic order, distinct from fast electronic cooling. Developed a 4-spin LLG model (Python) validating the spin-phonon coupling mechanism behind the signal.
- Performed DFT/BSE calculations (Quantum ESPRESSO, OCEAN, XSpectra) on monolayer 1T-VSe₂, characterizing the Stoner ferromagnetic instability to interpret pump-probe XUV spectra.

ESPCI Paris – PSL | PSL University

Jan. 2026 – Apr. 2026

Research Intern – LPEM Research Group | Advisor: Cheryl Feuillet-Palma

Paris, France

- Fabricated and characterized 15 nm Nb thin films (Hall cross geometry, Mn⁺-doped at 10¹²-10¹⁴ ions/cm²) via cryogenic transport measurements, resolving IV hysteresis with the Skocpol-Beasley-Tinkham self-heating model and identifying distinct critical-current pinning mechanisms across doping levels.
- Applied Ginzburg-Landau and Abrikosov-Gorkov theories to interpret critical temperature suppression across doping levels and analyzed the influence of controlled disorder geometry.

NTT Basic Research Laboratories (NTT BRL)

July 2025 – Dec. 2025

R&D Intern – Nanodevices Research Group | Advisor: Gento Yamahata

Atsugi, Japan

- Characterized silicon single-electron and -hole tunable-barrier pumps from room temperature to 60 mK (dilution refrigerator), achieving well-resolved $1ef/2ef$ quantized plateaus on the same quantum dot at back-gate biases beyond prior published results.
- Extracted transition temperatures benchmarking hole vs. electron pumping accuracy against effective-mass theory. Built a master-equation simulation (Python) identifying the optimal device parameter regime for minimizing pumping error.

Sorbonne University – French Chemistry Tournament (TFChim)

Feb. 2024 – June 2025

Research Team Member – ESPCI Team | Advisor: Yvette Tran

Paris, France

- Member of the 2025 winning ESPCI team in a highly competitive national competition against 10 top school such as ENS, École Polytechnique and Sorbonne University, involving oral defence of original research in chemistry.
- Led electrochemical synthesis of self-similar zinc dendrites (Zn²⁺/Cu redox deposition), characterizing fractal dimension via Minkowski-Bouligand box-counting and identifying concentration-dependent volumetric growth.
- Designed a multi-step organic synthesis route (Corey-Fuchs, Suzuki-Miyaura, hydroboration couplings) toward a self-similar tri-benzene mimicking a Sierpiński triangle, targeting applications in molecular magnetic confinement.

SELECTED PRESENTATIONS & REPORTS

Non-equilibrium dynamics in 2D van der Waals quantum materials: sHHG Spectroscopy of NiPS₃ & Ab Initio XUV Spectroscopy of 1T-VSe₂ | Zuerch Lab. (July 2026)
Competition between magnetic impurities and superconducting order in Nb thin films | LPEM Research Group. (Apr. 2026)
Qiskit Simulations | ESPCI Quantum Engineering module. (Apr. 2026)
Comparative Single-Charge Pumping in Silicon: Electron vs. Hole Performance in a Single Quantum Dot | NTT BRL. (Dec. 2025)
Numerical Simulation of the Quantum Harmonic Oscillator | ESPCI Quantum Physics module. (May 2024)

TECHNICAL SKILLS

Computation: Quantum ESPRESSO, OCEAN (BSE), XSpectra, PySCF, Qiskit, QuTiP.
Data & tools: Python (NumPy/SciPy), MATLAB, Igor Pro, Klayout, Vesta, ImageJ.
General: Git, LaTeX, SQL, Excel.

HONORS & AWARDS

ESPCI Alumni – Nominee for the Best Industrial Internship Award (2026).
EPFL Master Excellence Fellowship – 20 000 CHF/year and guaranteed housing (2026-2028).
French Chemistry Tournament (TFChim) – Winning Team Member (2025).
Joliot ESPCI Scholarship – 3 071€ (2023-2025), 1 535€ (2026).
French Government CROUS Scholarship – 2597€ (2021), 2701€ (2022), 3071€ (2023-2024), 2163€ (2025-2026).
French Government Merit-Based Baccalaureate Grant – 900€ (2021).

TEACHING EXPERIENCE

Academia | Terraskola | Prép’Assist Nov. 2023 – Present
Oral Examiner and Undergraduate-level Tutor in Maths, Physics and Chemistry Paris, France
• Prepared 11 students for national exams to top French engineering schools through one-on-one weekly tailored sessions.

French Ministry of Education Sept. 2024 – June 2025
Oral Examiner in Physics & Chemistry Paris, France
• Assessed 140+ students in real-exam oral conditions (6 hours/week) within France’s most selective pre-engineering courses at Lycée Louis-Le-Grand (LLG) and ENCPB (Lycée Pierre-Gilles de Gennes). Provided targeted scientific feedback.

Ambition Science ParisTech ENCPB | Climbing Together to Success Oct. 2023 – June 2024
Undergraduate-level Tutor in Maths and Physics Paris, France
• Mentored 4 students through tailored group sessions. Promoted social diversity and inclusion by actively supporting students from underrepresented backgrounds and helping them navigate through academic and administrative challenges.

WORK EXPERIENCE

Commission des Titres d’Ingénieur (CTI) Oct. 2024 – Oct. 2026
Engineering Student Expert Paris, France
• Audited French engineering schools alongside field experts. Contributed to official accreditation reports (Bologna Process).

Physique-Chimie Avenir (PCA) Mar. 2024 – Mar. 2025
Quality Manager | ESPCI Junior Enterprise (L30) Paris, France
• Managed quality assurance across 20 client studies (Bouygues, Solvay, EDF, Veolia; 50 k€ revenue). Implemented document management and archiving workflows in line with CNJE standards.
• Served as a board member and contributed to strategic decision-making with the presidency’s action plan.

Forum Horizon Chimie (FHC) Oct. 2023 – Oct. 2024
Digital Manager | Event Organizer and Mediator | 38th edition Paris, France
• Led the redesign/overhaul of the Forum Horizon Chimie website and social media. Coordinated event stewards/mediators and managed visitor operations.

CERTIFICATIONS

IBM Quantum | Qiskit Global Summer School 2026 – Quantum Excellence Badge. [Earned in July 2026]
TOEFL iBT – Score: 104/120. [Earned in Nov. 2025]
IBM Quantum – Basics in Quantum Information & Fundamentals of Quantum Algorithms. [Earned in Oct. 2025]